Modeling the Effect of a Distractor on Tactile Categorization

Models A, B and C

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TODO: date and venue

Outline

Task and Data

Model A: Sessions Pooled

Model B: Sessions Hierarchical

Model C: Distractor Amplitude as a Covariate

Summary

The task

- The monkey feels a vibration on one finger and reports whether it is "high" or "low" relative to a trained boundary of 28 μm.
- Unimanual trials: stimulus only. Bimanual trials: a distractor vibration on the other hand at the same time.
- Question: how does the distractor change the categorization?

TODO: schematic of a trial (stimulus hand, distractor hand, response)

The data

- One monkey (Sirius), 42 sessions, 12,937 trials.
- 8 stimulus amplitudes, 6–50 μm; 4 groups: hand $\times$ (unimanual, bimanual).
- Each session uses a single distractor amplitude (6–32 μm) on its bimanual trials.

TODO: figure: raw proportions of "high" by amplitude and group

Psychometric function

Every model describes $P(\text{“high”})$ at stimulus x (standardized amplitude) by

$$\psi(x) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x)$$

- γ_h, γ_l: lapse rates (“high” to the weakest, “low” to the strongest stimuli)
- Summaries: PSE (where $\psi = 0.5$) and JND (half the distance between $\psi = 0.25$ and 0.75), in μm

TODO: one example curve with PSE and JND marked

Three models

Model	Sessions	Distractor
A	pooled: one curve per group	uni and bi fit separately
B	hierarchical: one curve per group and session	uni and bi fit separately
C	hierarchical, one group per hand	covariate: shift $\beta_2 d$

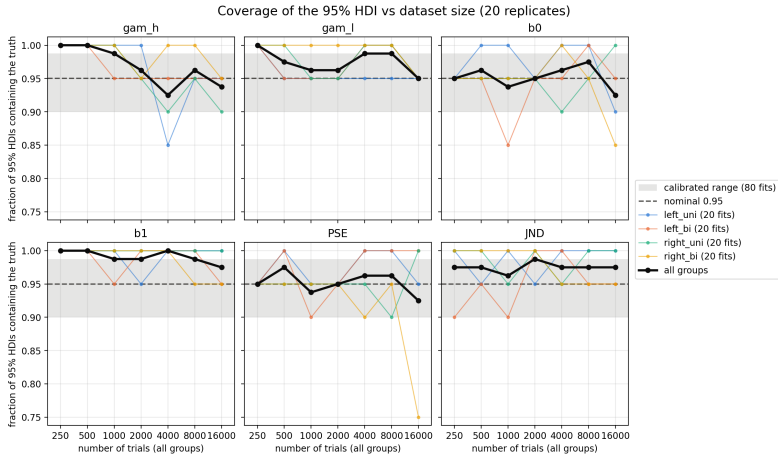
For each model: convergence study on synthetic data, then the fit to the real data.

Model A

- One psychometric curve per group (hand $\times$ condition), shared by all 42 sessions: 16 parameters.
- Priors: $\beta_0 \sim \mathcal{N}(0, 6)$, $\beta_1 \sim \mathcal{N}(4, 2)$, $\gamma = 0.25\, w$ with $w \sim$ Beta.

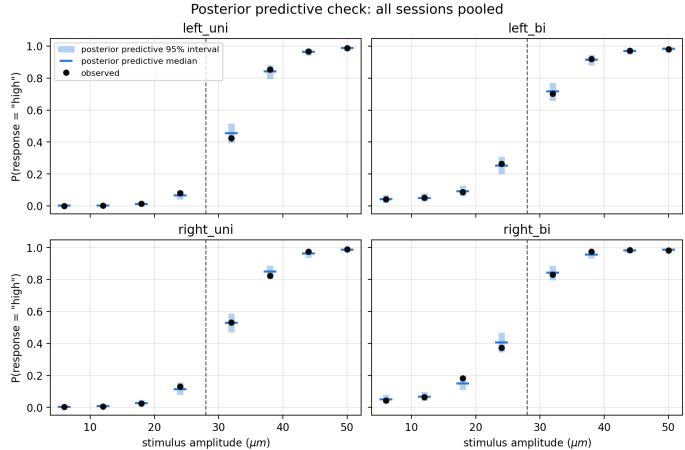
TODO: one-line motivation: simplest model, a baseline

Model A: convergence on synthetic data



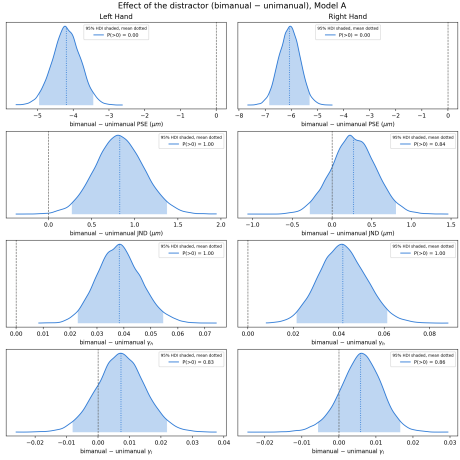
TODO: takeaway; numbers from Convergence_A_Report, "Conclusions"

Model A: fit to the real data



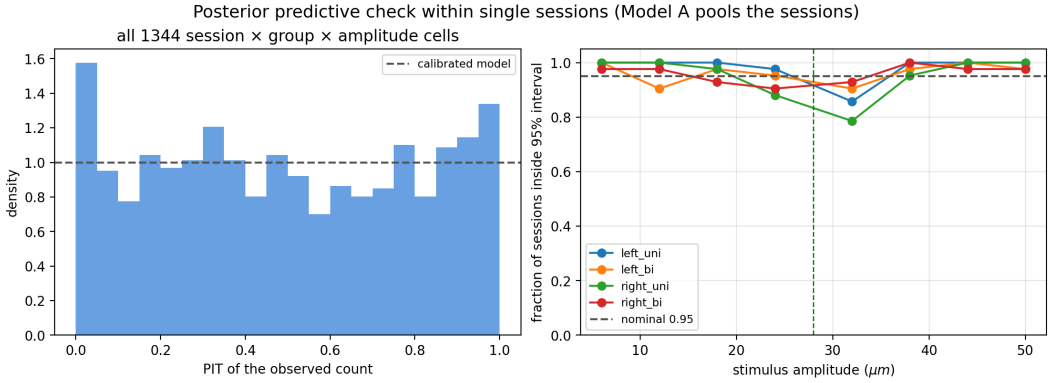
TODO: pooled check passes: all 32 proportions inside (Fit_A_Report, "Model Checks")

Model A: effect of the distractor



TODO: PSE shift, γ_h and JND effects (Fit_A_Report, Summary)

Model A misses the variation between sessions



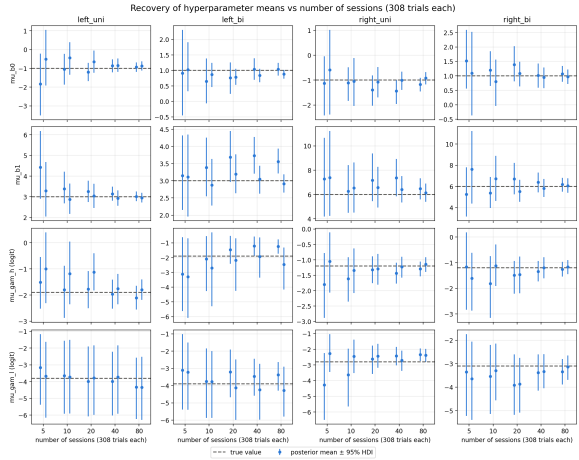
TODO: within single sessions responses vary more than A predicts (Fit_A_Report, "Model Checks")

Model B

- A curve for every group *and session*, drawn from group-level distributions (means and between-session sds).
- Design-based priors, checked by prior predictive simulation.

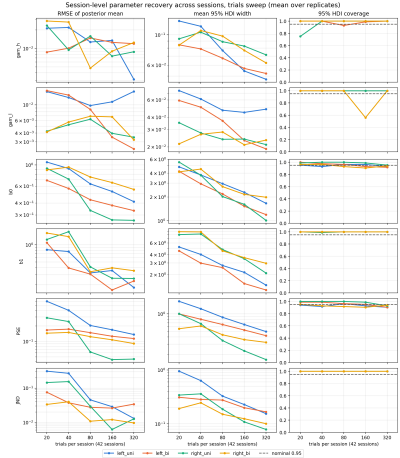
TODO: plate diagram or equations (Convergence_B_Report, "Model Structure")

Model B: convergence on synthetic data



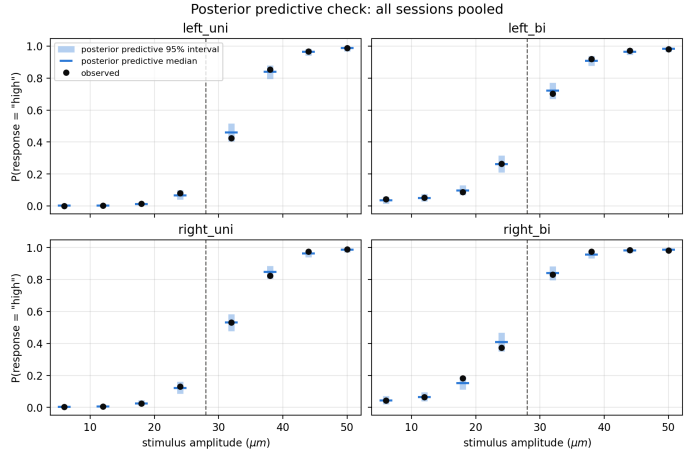
TODO: what is recovered at the real data size (Convergence_B_Report, "Conclusions")

Model B: per-session recovery



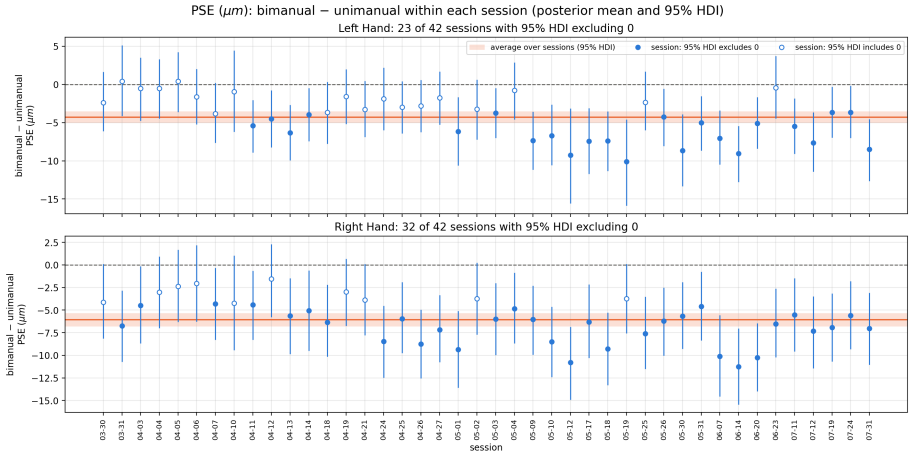
TODO: per-session PSE and JND reliable at ∼300 trials

Model B: fit to the real data



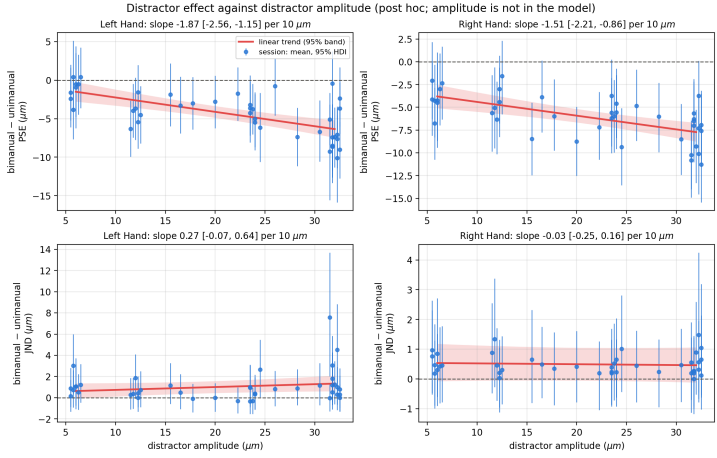
TODO: all 32 proportions inside; 99.8% of session cells (Fit_B_Report, "Posterior predictive checks")

Model B: PSE shift in every session



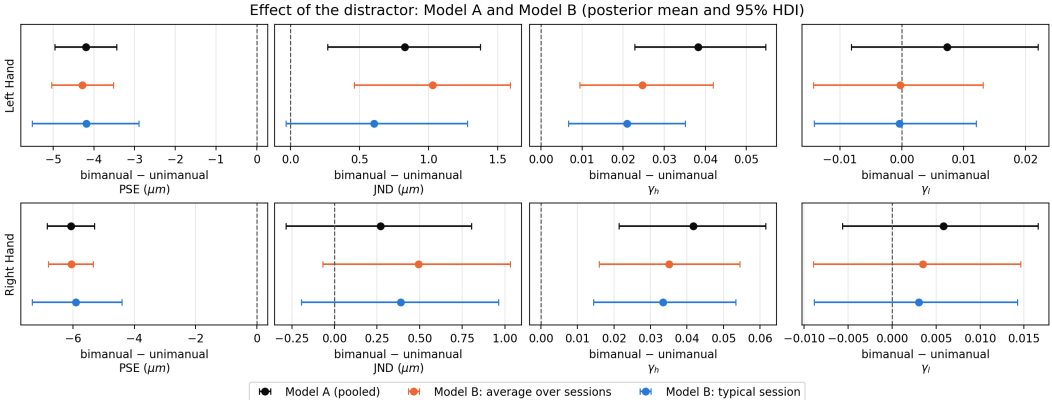
TODO: shift negative in 40/42 (left) and 42/42 (right) sessions (Fit_B_Report, Summary)

Model B: the shift grows with distractor amplitude (exploratory)



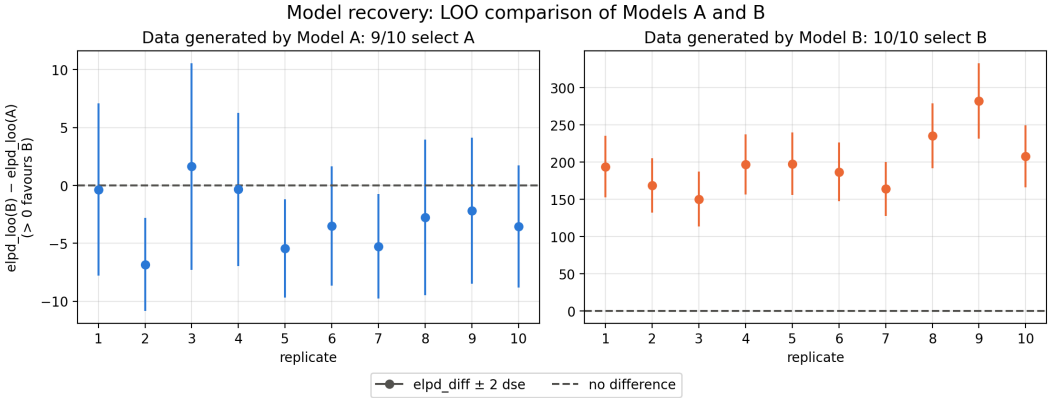
TODO: about -2 μm per 10 μm; confounded with session order

Model B against Model A



TODO: same effects; B better by LOO, 164 (SE 19) (Fit_B_Report, "Comparison with Model A")

Is B's advantage real? Model recovery



TODO: B wins on all data from B, ties or loses on data from A (Model_Recov_BA_Report)

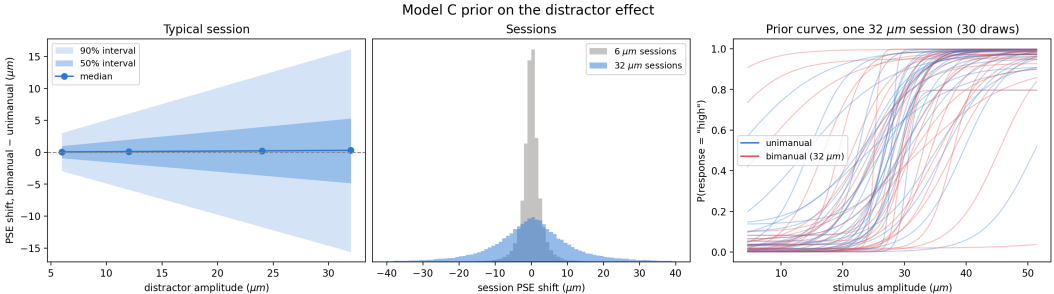
Model C

$$\psi(x, d) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x + \beta_2 d)$$

- d: distractor amplitude (0 on unimanual trials); one group per hand.
- Uni- and bimanual trials share slope and lapse rates: the distractor only shifts the curve, by an amount proportional to d.

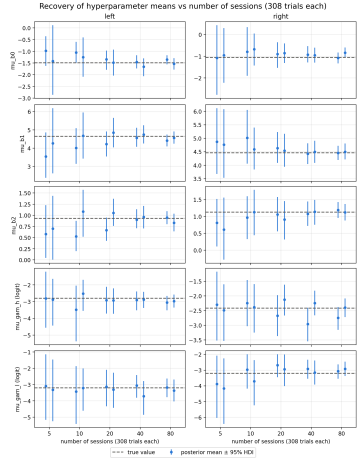
TODO: why: test the amplitude trend from Model B inside a model

Model C: prior on the distractor effect



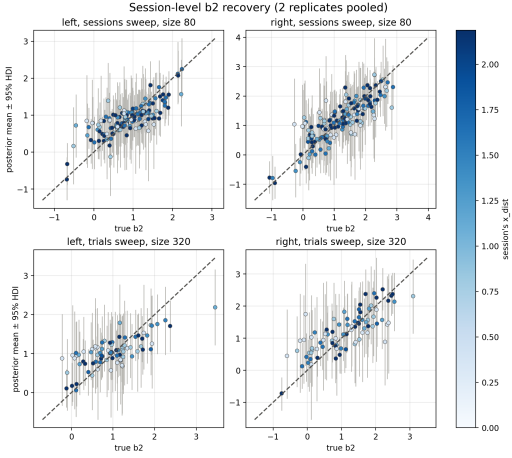
TODO: weakly informative; a weak distractor can only shift a little (Convergence_C_Report)

Model C: convergence on synthetic data



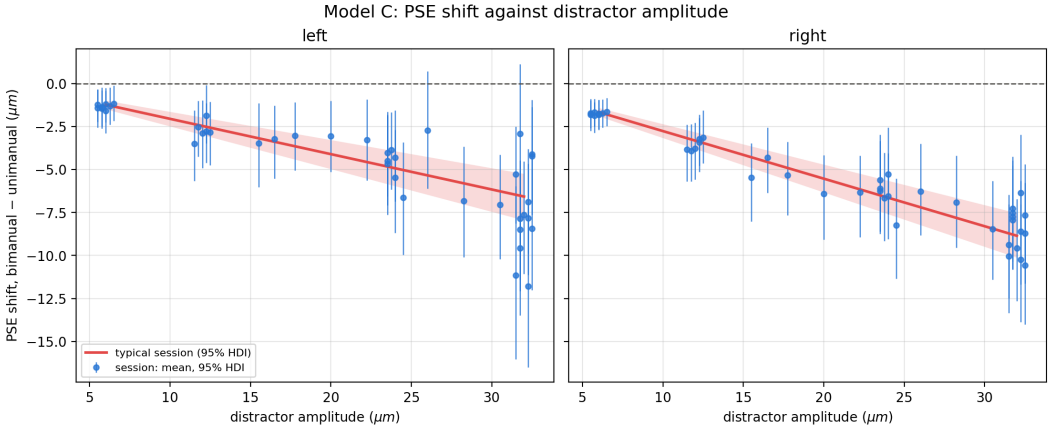
TODO: all hand-level means recovered, μ_{β_2} well (Convergence_C_Report, "Conclusions")

Model C: per-session β_2 is weak for small distractors



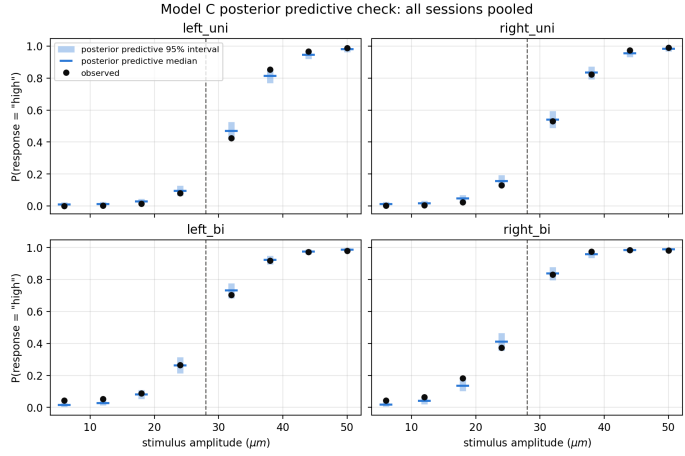
TODO: shrinkage toward the hand mean at 6 μm

Model C: the shift grows with distractor amplitude



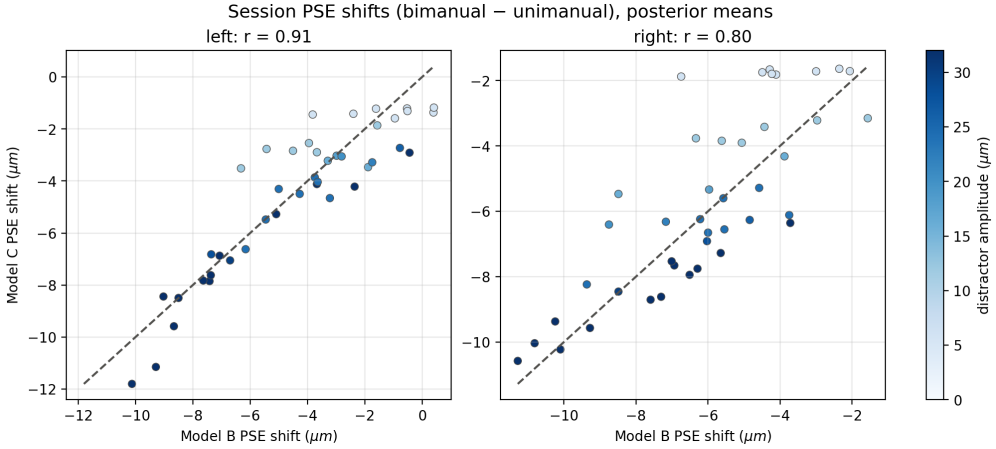
TODO: μ_{β_2} and the typical shift at 6 and 32 μm (Fit_C_Report, "Effect of the Distractor")

Model C misfits at the weakest stimuli



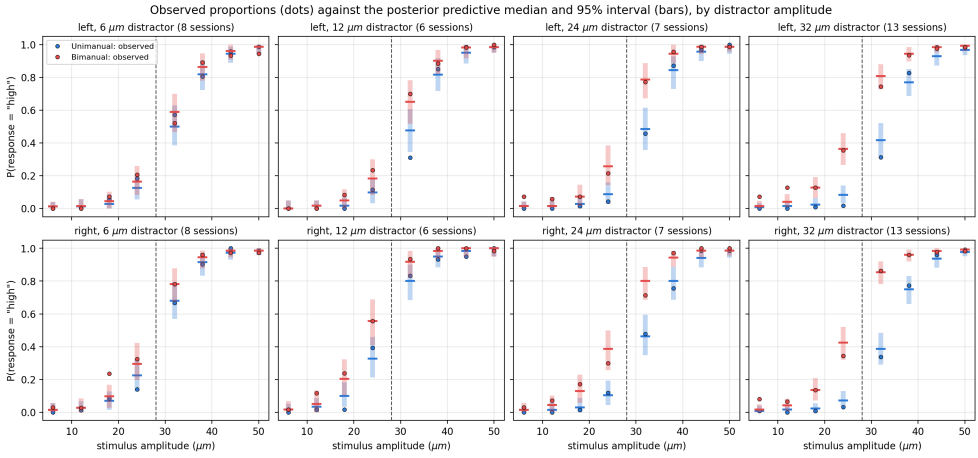
TODO: shared lapse rates: 6 of 32 proportions outside (Fit_C_Report, "Posterior predictive check")

Model C against Model B



TODO: LOO favors B by 37 (SE 12); 6 μm shifts too small (Fit_C_Report, Comparison)

Model C: shift by distractor amplitude, observed vs predicted



TODO: right hand at 6 μm outside: a presence effect (Fit_C_Report, amplitude check)

Comparison of the models

	A	B	C
Sessions	pooled	hierarchical	hierarchical
Distractor	separate groups	separate groups	covariate $\beta_2 d$
elpd_{loo}	**TODO:**	**TODO:**	**TODO:**
PSE shift, left / right (μm)	**TODO:**	**TODO:**	**TODO:**

TODO: fill from Fit_A/B/C_Report; elpd differences from Fit_B and Fit_C

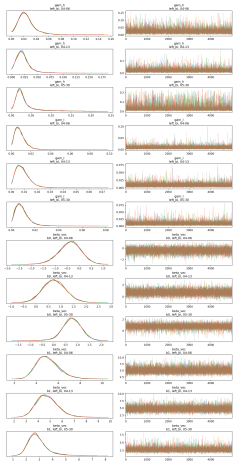
Conclusions

- The distractor lowers the PSE: **TODO:** size, both hands.
- It raises the rate of "high" responses to weak stimuli (γ_h).
- The sessions differ; a hierarchical model is needed.
- The shift grows with distractor amplitude, **TODO:** but not proportionally?

Next steps

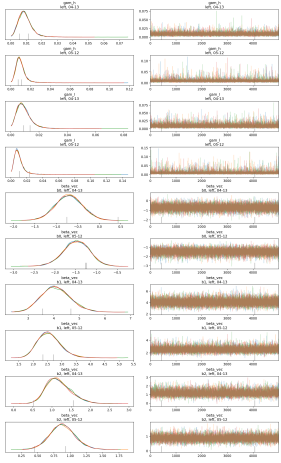
- Revise Model C: lapse rates per condition, a distractor-presence term, possibly a bimanual slope.
- Model recovery for B against the revised C.
- **TODO:** further data or experiments

Backup: Model B sampling diagnostics



TODO: no divergences, $\hat{R} = 1.00$ (Fit_B_Report, "Sampling diagnostics")

Backup: Model C sampling diagnostics



TODO: 2 divergences in 20,000 draws, $\hat{R} = 1.000$ (Fit_C_Report)

Backup: Model C session PSEs

